



Rui Casaca

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About me

Computer Engineering graduate from ISEC with hands-on experience in modern software development, including React, TypeScript, Java, and Python, gained during an internship at Critical Software. I bring 15 years of professional experience, discipline, and a strong commitment to continuous learning and adapting to new technologies. Motivated by solving real problems through reliable, well-designed software, and open to different areas within software engineering, from application development to systems and platforms.

Technical Skills

Languages: TypeScript, Python, Java, C/C++, C#

Frontend: React, HTML/CSS

Mobile: Flutter, Kotlin

Backend: Node.js

Familiar With: C#, ASP.NET Core (.NET 8), Entity Framework Core

Databases: SQLite, MongoDB, Firebase

Tools: Git, GitHub, BitBucket, Docker, Postman, Confluence

Professional Experience

Software Engineering Intern *Critical Software*

Feb 2026 - Jul 2026

Space Systems: Ground Segment, Telemetry, ECSS, CCSDS Space Packets, Software Traceability, System Validation

- Contributed to the development of Lighthouse, a Ground Segment platform for space missions.
- Worked on software features for real-time telemetry visualization and processing, integrating REST APIs and WebSocket services.
- Gained exposure to ECSS standards, spacecraft telemetry concepts, traceability practices, and CCSDS space packet structures while collaborating in a space systems engineering environment.

Freelance GIS Specialist & 3D Modeling *Fiverr, upwork*

Nov 2022 - Present

- Designed and delivered GIS solutions, data-driven applications and 3D modelling projects for clients in mission-critical contexts, translating technical and operational requirements into robust, usable tools.
- Coordinated small teams (2-5 people) and ensured on-time delivery and quality standards across multiple projects.

Military & GIS Analyst and Operations Technician

GNR (National Republican Guard)

Oct 2011 - Mar 2019

UEPS (Emergency Protection & Relief Unit)

Mar 2019 - Feb 2026

- Supported missing persons search and disaster response operations through the use of drones and Geographic Information Systems (GIS).
- Worked in mission-critical environments requiring rapid decision-making, teamwork and operational accuracy.

Projects

HomeVault - Offline-First Inventory Management App

Flutter, Riverpod, Drift (SQLite), Google Drive API

- Developed an Android home inventory management application with an offline-first architecture, enabling users to catalog assets, documents and photos securely.
- Implemented local data management with Drift (SQLite), encrypted backups and Google Drive synchronization, biometric authentication, QR/barcode scanning and Excel export.

Missing Persons Search Application

React, TypeScript, Node.js, LLM APIs

- Developed a full-stack platform supporting missing persons search operations, enabling centralized information management and analysis. Integrated LLM APIs for contextual data analysis, implemented real-time dashboards and automated PDF report generation with email distribution.

Teacher and Student Evaluation Platform (Academic)

Java, Client-Server Architecture, RMI, SQLite

- Developed a client-server application for teacher and student evaluation processes, including authentication, role-based access control, questionnaire management, result processing and SQLite-based data storage.

Education

Bachelor's in Computer Engineering – Software Development Track

Instituto Superior de Eng. de Coimbra (ISEC)

2023 - 2026